Survey of Math

MA-105 Worcester State University

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Instructor Lastname
University of Templates

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Contents

Chapter 1

Course Documents

Syllabus

Course Information

This is the syllabus for **course name (MATH xxx, section xxx)** for [term] 20xx. It is a [n] credit course.

Instructor	Prof. Lastname, Office Location, prof.lastname@example.edu.
Student Hours	TBD
Class meets	course times and location.
Course Description	course description from catalog
Prerequisite	list of prerequisites
Textbook and course materials	*textbook name* by textbook author.

Assessments and Grades

Chapter 2

Voting and Apportionment

Beyond just marking a ballot, these concepts shape how our voices are heard and how resources are distributed. This chapter breaks down the mechanics behind decision-making, offering clear examples of how mathematical fairness directly impacts your community and representation.

Voting methods

How to determine a Winner?

> **Preference ballots** are ballots in which a voter is asked to rank all candidates in order of preference. A **preference table** shows how often each particular outcomes occurred.

Example 2.1 Four candidates are running for student body president: Alice (A), Bonnie (B), Carlos (C) and Daniel (D). The students were asked to rank all the candidates in order of preference.

Table 2.2 Preference Table For The Election Of Student Body President

Number of Votes	130	120	100	150
First choice	A	D	D	C
Second choice	B	B	B	B
Third choice	C	C	A	A
Fourth choice	D	A	C	D

1. How many students voted in the election?
2. How many students selected the candidates in this order : D, B, A, C?
3. How many students selected Daniel (D) as their first choice for student body president?

Solution.

1. To find the total number of students, sum the votes in the top row: $130 + 120 + 100 + 150 = 500$ students.
2. Looking at the columns, the order ***D, B, A, C*** appears in the third column. There were 100 students who voted in this order.

3. Daniel (D) is listed as the first choice in the second and third columns. The total number of students is $120 + 100 = 220$ students.

▲

Definition 2.3 The Plurality Method. The candidate (or candidates, if there is more than one) with the most first-place votes is the winner. ◆

Example 2.4 Table 2.2 shows the preference table for the four candidates running for student body president. Who is declared the winner using the Plurality method?

Solution. A received 130 first-place votes, D received $120 + 100 = 220$ first-place votes and C received 150 first-place votes. Thus Daniel is the winner. ▲

Majority Rule Decision is when more than 50% of the voters rank a candidate in first-place. The majority rule will almost always have a winner.

Definition 2.5 The Plurality-with-Elimination Method. The candidate with the majority (over 50% votes) of first-place votes is the winner. If no candidate receives a majority, eliminate the candidate with the fewest first-place votes. Move the candidates below them up one place and recount. Repeat until a candidate has a majority. ◆

Example 2.6 Elimination Method Example. Determine the winner for the student president election from Table 2.2, using Plurality-with-Elimination.

Solution. Total votes = 500. Majority needed = 251.

Round 1: A=130, D=220, C=150, B=0. No majority. Bonnie (B) is eliminated.

Table 2.7 Preference Table: Round 2 (Bonnie Eliminated)

Number of Votes	130	120	100	150
First choice	A	D	D	C
Second choice	C	C	A	A
Third choice	D	A	C	D

Round 2: After shifting votes up, the first-place votes remain: A=130, D=220, C=150. Still no majority. Alice (A) has the fewest (130) and is eliminated.

Table 2.8 Preference Table: Final Round (Alice and Bonnie Eliminated)

Number of Votes	130	120	100	150
First choice	C	D	D	C
Second choice	D	C	C	D

Round 3: Alice's 130 votes go to her next preferred candidate, Carlos (C). New totals: Daniel = 220, Carlos = $150 + 130 = 280$.

Carlos now has 280 votes (a majority). Carlos is the winner. ▲

Definition 2.9 The Borda Count Method. Each voter ranks the candidates from the most favorable to the least favorable. Each last-place vote is given 1 point, each next-to-last-place vote is given 2 points, and so on. The points are totaled for each candidate separately. The candidate with the most points is the winner. ◆

Example 2.10 Borda Count Example. Using the preference data from Table 2.2, who is declared the winner using the Borda Count Method?

Solution. With 4 candidates, 1st place gets 4 pts, 2nd gets 3 pts, 3rd gets 2 pts, and 4th gets 1 pt.

Table 2.11 Borda Count Calculation Table

Number of Votes	130	120	100	150
1st choice: 4 pts	A: $130 \times 4 = 520$	D: $120 \times 4 = 480$	D: $100 \times 4 = 400$	C: $150 \times 4 = 600$
2nd choice: 3 pts	B: $130 \times 3 = 390$	B: $120 \times 3 = 360$	B: $100 \times 3 = 300$	B: $150 \times 3 = 450$
3rd choice: 2 pts	C: $130 \times 2 = 260$	C: $120 \times 2 = 240$	A: $100 \times 2 = 200$	A: $150 \times 2 = 300$
4th choice: 1 pt	D: $130 \times 1 = 130$	A: $120 \times 1 = 120$	C: $100 \times 1 = 100$	D: $150 \times 1 = 150$

- ***Alice (A):*** $(130 \times 4) + (120 \times 1) + (100 \times 2) + (150 \times 2) = 520 + 120 + 200 + 300 = 1140$ pts
- ***Bonnie (B):*** $(130 \times 3) + (120 \times 3) + (100 \times 3) + (150 \times 3) = 390 + 360 + 300 + 450 = 1500$ pts
- ***Carlos (C):*** $(130 \times 2) + (120 \times 2) + (100 \times 1) + (150 \times 4) = 260 + 240 + 100 + 600 = 1200$ pts
- ***Daniel (D):*** $(130 \times 1) + (120 \times 4) + (100 \times 4) + (150 \times 1) = 130 + 480 + 400 + 150 = 1160$ pts

Bonnie is the winner with 1500 points. ▲

Definition 2.12 The Pairwise Comparison Method. Every candidate is matched on a one-to-one basis with every other candidate. If candidate A beats candidate B in one-to-one competition, then A receives 1 point and B receives 0. If they tie, each receives 0.5 points. The candidate with the most total points is the winner.

In an election with n candidates, the number of comparisons, C, is calculated as:

$$C = \frac{n(n-1)}{2}$$

◆

Example 2.13 Pairwise Comparison Example. Using the same preference Table 2.2, for the four candidates (Alice, Bonnie, Carlos, and Daniel), who is the winner using the Pairwise Comparison Method?

Solution. With $n = 4$ candidates, there are $\frac{4(3)}{2} = 6$ total comparisons (A vs.B, A vs.C, A vs.D, B vs.C, B vs.D, C vs.D) and determine the winner using the pairwise comparison method.

130	120	100	150
A	B	B	B
B	A	A	A

A vs B — ***A***

130	120	100	150
A	C	A	C
C	A	C	A

A vs C — ***A***

130	120	100	150
A	D	D	A
D	A	A	D

A vs D — ***A***

130	120	100	150
B	B	B	C
C	C	C	B

B vs C — ***B***

130	120	100	150
B	D	D	B
D	B	B	D

B vs D — ***B***

130	120	100	150
C	D	D	C
D	C	C	D

C vs D — ***C***

1. ***A vs B:*** Alice is preferred over Bonnie in 130 ballots. Bonnie is preferred over Alice in $120 + 100 + 150 = 370$ ballots. **Winner: Bonnie (B gets 1 pt)**
2. ***A vs C:*** Alice is preferred over Carlos in $130 + 120 + 100 = 350$ ballots.

Carlos is preferred over Alice in 150. **Winner: Alice (A gets 1 pt)**

3. ***A vs D:*** Alice is preferred over Daniel in 130 ballots. Daniel is preferred over Alice in $120 + 100 + 150 = 370$ ballots. **Winner: Daniel (D gets 1 pt)**

4. ***B vs C:*** Bonnie is preferred over Carlos in $130 + 120 + 100 = 350$ ballots. Carlos is preferred over Bonnie in 150. **Winner: Bonnie (B gets 1 pt)**

5. ***B vs D:*** Bonnie is preferred over Daniel in $130 + 150 = 280$ ballots. Daniel is preferred over Bonnie in $120 + 100 = 220$. **Winner: Bonnie (B gets 1 pt)**

6. ***C vs D:*** Carlos is preferred over Daniel in $130+150 = 280$ ballots. Daniel is preferred over Carlos in $120 + 100 = 220$. **Winner: Carlos (C gets 1 pt)**

Final Point Totals:

- Bonnie (B): 3 points
- Alice (A): 1 point
- Daniel (D): 1 point
- Carlos (C): 1 point

Bonnie is the winner using Pairwise Comparison. ▲

> If a candidate wins all pair comparisons with other candidates, that candidate is called the **Condorcet winner**. A Condorcet winner automatically wins by the pairwise comparisons method.

Example 2.14 Condorcet Winner. Is there a Condorcet winner in the following election?

Table 2.15 Preference Table for Condorcet Example

Number of Votes	30	30	30	20
First choice	B	B	A	A
Second choice	A	A	C	D
Third choice	C	D	D	C
Fourth choice	D	C	B	B

Solution. To find a Condorcet winner, we must see if one candidate wins every head-to-head matchup. Let's check candidate ***B*** and candidate ***A***:

- ***B vs A:*** B is ranked higher than A in $30 + 30 = 60$ ballots. A is ranked higher than B in $30 + 20 = 50$ ballots. **Winner: B**.
- ***B vs C:*** B is ranked higher than C in $30 + 30 + 30 + 20 = 110$ ballots (every ballot). **Winner: B**.
- ***B vs D:*** B is ranked higher than D in $30 + 30 + 30 + 20 = 110$ ballots (every ballot). **Winner: B**.

Since ***B*** beats every other candidate in a one-on-one comparison, ***B is the Condorcet winner***. ▲

Arrow's Impossibility Theorem states that a ranked-voting electoral system cannot reach a community-wide ranked preference by converting individuals' preferences while meeting all the conditions of a fair voting system.

Consider the following situation with 3 candidates in one-to-one competition (pairwise comparison method):

- Suppose candidate A beats candidate B
- Suppose candidate B beats candidate C
- Suppose candidate C beats candidate A

When this occurs, it is because the conflicting majorities are each made up of different groups of individuals. No winner can be determined.

This is called the **Condorcet Paradox** (also known as the voting paradox).

Example 2.16 Condorcet Paradox. Who would win this election through the pairwise comparison method?

Table 2.17 Preference Table: Three-Candidate Election

Number of Votes	8	9	10
First choice	C	B	A
Second choice	A	C	B
Third choice	B	A	C

Solution. We perform the three possible head-to-head comparisons:

- ***A vs B:*** A is preferred on 8 (col 1) + 10 (col 3) = 18 ballots. B is preferred on 9 (col 2) ballots. **Winner: A** (A gets 1 pt).
- ***B vs C:*** B is preferred on 9 (col 2) + 10 (col 3) = 19 ballots. C is preferred on 8 (col 1) ballots. **Winner: B** (B gets 1 pt).
- ***C vs A:*** C is preferred on 8 (col 1) + 9 (col 2) = 17 ballots. A is preferred on 10 (col 3) ballots. **Winner: C** (C gets 1 pt).

Each candidate (A, B, and C) has exactly 1 point. This is a **Condorcet Paradox** because there is a "rock-paper-scissors" loop (A beats B, B beats C, and C beats A), resulting in a three-way tie. ▲

Checkpoint 2.18 Mayor of a Small Town. Three candidates A, B, and C are running for mayor of a small town. The results of the election are shown in the following preference table.

Table 2.19 Mayor Election Preference Table

Number of Votes	1200	900	900	600
First choice	A	C	B	B
Second choice	B	A	C	A
Third choice	C	B	A	C

1. Determine the winner using the Plurality method.
2. Determine the winner using the Borda count method.
3. Determine the winner using the Plurality-with-elimination method.

4. Is there a Condorcet Winner?

Answer.

1. ***Plurality:*** A = 1200, B = $900 + 600 = 1500$, C = 900. **Winner: B**.
2. ***Borda Count (3-2-1):***
   - A: $(1200 \times 3) + (900 \times 2) + (900 \times 1) + (600 \times 2) = 3600 + 1800 + 900 + 1200 = 7500$
   - B: $(1200 \times 2) + (900 \times 1) + (900 \times 3) + (600 \times 3) = 2400 + 900 + 2700 + 1800 = 7800$
   - C: $(1200 \times 1) + (900 \times 3) + (900 \times 2) + (600 \times 1) = 1200 + 2700 + 1800 + 600 = 6300$

   Winner: B.
3. ***Plurality-with-Elimination:*** Total votes = 3600; Majority = 1801. Round 1: A=1200, B=1500, C=900. No majority. Eliminate C. C's 900 votes go to A (second choice). Final: A = $1200 + 900 = 2100$, B = 1500. **Winner: A**.
4. ***Condorcet Winner Check:***
   - A vs B: A is preferred on $1200 + 900 = 2100$. B on 1500. (A wins)
   - A vs C: A is preferred on $1200 + 600 = 1800$. C on $900 + 900 = 1800$. (Tie)

   Since A does not win all matches, **No Condorcet Winner exists**.

Solution.

1. *Plurality Method.*

   We count only the first-place votes for each candidate:
   - ***A:*** 1200 votes
   - ***B:*** $900 + 600 = 1500$ votes
   - ***C:*** 900 votes

   Winner: B (1500 votes).
2. *Borda Count Method (3-2-1 Points).*

   Table 2.20 Borda Count Point Distribution

Candidate	1st (3 pts)	2nd (2 pts)	3rd (1 pt)	Total Poin
A	$1200 \times 3 = 3600$	$(900 + 600) \times 2 = 3000$	$900 \times 1 = 900$	7500
B	$(900 + 600) \times 3 = 4500$	$1200 \times 2 = 2400$	$900 \times 1 = 900$	7800
C	$900 \times 3 = 2700$	$900 \times 2 = 1800$	$(1200 + 600) \times 1 = 1800$	6300

   Winner: B (7800 points).
3. *Plurality-with-Elimination.*

   Total Votes: 3600. Majority needed: 1801.

   Round 1: A=1200, B=1500, C=900. No candidate has 1801. Eliminate C.

Table 2.21 Final Round (C Eliminated)

Votes	1200	900	900	600
1st	A	A	B	B
2nd	B	B	A	A

New totals: $\boldsymbol{A} = 1200 + 900 = 2100$. $\boldsymbol{B} = 900 + 600 = 1500$. **Winner: A** (2100 votes).

Checkpoint 2.22 Math Club Presidential Election. Four candidates are running for president of the Math Club: Paula (P), Sylvia (S), Craig (C), and Brian (B). The results of the election are shown in the following preference table:

Table 2.23 Election Preference Schedule

Number of Votes	15	19	23	10	18	15
First choice	B	C	P	P	S	B
Second choice	S	P	S	B	C	S
Third choice	P	S	B	C	P	C
Fourth choice	C	B	C	S	B	P

1 How many students voted in the election?

2 How many students voted Brian as their first choice?

3 Who would win the presidency using the Plurality method?

4 Is there a Majority Winner?

5 Who would win the presidency using the Plurality with Elimination method?

6 Who would win the presidency using the Borda count method?

7 Who would win the presidency using the pairwise comparison method? If a tie, use the Plurality method between the winners to determine the tie-breaker.

Answer.

1 100 students

2 30 students

3 Paula (33 votes)

4 No (Majority requires 51 votes)

5 Craig

6 Sylvia (279 points)

7 Paula (3 points)

Solution.

1 Total Votes: 100.

2 Brian 1st choice: 30.

3 Plurality Winner: Paula (33).

4 Majority: No (Requires 51).

5 Plurality with Elimination:

Round 1: Sylvia has the fewest 1st-place votes (18) and is eliminated.

Table 2.24 Round 2 (Sylvia Eliminated)

Votes	15	19	23	10	18	15
1st	B	C	P	P	C	B
2nd	P	P	B	B	P	C
3rd	C	B	C	C	B	P

New totals: P=33, B=30, C=37 (19 + 18). Brian is eliminated.

Table 2.25 Round 3 (Brian Eliminated)

Votes	15	19	23	10	18	15
1st	P	C	P	P	C	C
2nd	C	P	C	C	P	P

Round 3: Paula vs. Craig. Craig wins the final matchup against Paula 52 to 48.

Winner: Craig.

6 Borda Count Method:

Using 4 pts for 1st, 3 for 2nd, 2 for 3rd, and 1 for 4th:

- Paula: $33(4) + 19(3) + 33(2) + 15(1) = 270$
- Sylvia: $18(4) + 53(3) + 19(2) + 10(1) = 279$
- Craig: $19(4) + 18(3) + 25(2) + 38(1) = 218$
- Brian: $30(4) + 10(3) + 23(2) + 37(1) = 233$

Winner: Sylvia.

7 Pairwise Comparison:

For the **Pairwise Comparison Method**, we compare every possible pair. There are 100 total votes, so 51 are needed to win a matchup:

- P vs. S: P has 52; S has 48. (P gets 1 pt)
- P vs. C: P has 63; C has 37. (P gets 1 pt)
- P vs. B: P has 70; B has 30. (P gets 1 pt)
- S vs. C: S has 81; C has 19. (S gets 1 pt)
- S vs. B: S has 60; B has 40. (S gets 1 pt)
- C vs. B: C has 60; B has 40. (C gets 1 pt)

Total Points: Paula = 3, Sylvia = 2, Craig = 1, Brian = 0. Paula is the winner; no tie-breaker is required.

Apportionment

Apportionment is the method of distributing power in political systems based on population density. Apportionment may reflect the availability of buses for city routes or doctors for patients.

Definition 2.26 The Standard Divisor. The standard divisor is found by dividing the total population under consideration by the number of items to be allocated.

$$\text{Standard divisor} = \frac{\text{total population in a group}}{\text{total number of items to be apportioned}}$$

◆

Definition 2.27 The Standard Quota. The standard quota for a particular group is found by dividing that group's population by the standard divisor.

$$\text{Standard quota} = \frac{\text{population of a particular group}}{\text{standard divisor}}$$

◆

Example 2.28 Palm Springs is composed of four states: A, B, C, and D. According to the country's constitution, the congress will have 30 seats, divided among the four states according to their respective populations. Find the standard quotas for states A, B, C, and D and complete the table below.

Table 2.29 Population of Palm Springs by State

State	**A**	**B**	**C**	**D**	**Total**
Population (in thousands)	275	383	465	767	1890
Standard Quota					

Solution. First, we find the **Standard Divisor** (SD):

$$SD = \frac{1890}{30} = 63$$

Now, divide each state's population by the divisor to find the **Standard Quota** (SQ):

- State A: $275/63 \approx 4.3651$
- State B: $383/63 \approx 6.0794$
- State C: $465/63 \approx 7.3810$
- State D: $767/63 \approx 12.1746$

Table 2.30

State	**A**	**B**	**C**	**D**	**Total**
Population (in thousands)	275	383	465	767	1890
Standard Quota	4.365	6.079	7.381	12.175	

▲

The **lower quota** is the standard quota rounded down to the nearest whole number. The **upper quota** is the standard quota rounded up to the nearest whole number.

A group's apportionment should be either its upper quota or its lower quota. An apportionment method that guarantees that this will always occurs is said to **satisfy the quota rule**.

Algorithm 2.31 Hamilton's Method.

1. *Calculate each group's standard quota.*

2. *Find the lower quota for each group.*

3. *Give surplus items, one at a time, to groups with the largest decimal parts until no surplus remains.*

Example 2.32 From Table 2.29, use Hamilton's method to apportion the 30 congressmen for Palm Springs.

Table 2.33 Hamilton's Method Table

State	**A**	**B**	**C**	**D**	**Total**
Population	275	383	465	767	1890
Standard Quota	4.365	6.079	7.381	12.175	30.00
Lower Quota	4	6	7	12	29
Surplus	+0	+0	+1	+0	1
Apportionment	4	6	8	12	30

1. The Standard Divisor is 63.
2. Lower quotas sum to 29, leaving 1 surplus seat.
3. State C has the largest decimal part (.381), so it receives the surplus seat.

Solution. Following the steps of **Hamilton's Method**:

1. The sum of the **Lower Quotas** is $4 + 6 + 7 + 12 = 29$. Since we must allocate 30 seats, there is $30 - 29 = 1$ surplus seat.

2. We look at the decimal parts of the Standard Quotas:

   - A: 0.365
   - B: 0.079
   - C: 0.381
   - D: 0.175

3. State C has the largest decimal part (0.381), so it receives the 1 surplus seat.

4. Final counts: A=4, B=6, C=8, D=12.

▲

Modified Divisor (MD).

A **Modified divisor** is a divisor, chosen by guess-and-check, that is used to make the sum of the modified quotas exactly equal to the number of items to be apportioned.

Algorithm 2.34 Jefferson's Method. *Find a* ***modified divisor*** *such that when each group's modified quota is rounded down (****modified lower quota****), the sum equals the number of items to be apportioned. Apportion to each group its modified lower quota.*

1. *Calculate the* ***Standard Divisor*** *(SD = Total Pop/Seats).*

2. *Calculate* ***Standard Quotas*** *and round down to find* ***Lower Quotas****.*

3. *If the sum of Lower Quotas is less than the total seats, choose a* ***Modified Divisor*** *(MD) that is* smaller *than the SD.*

4. *Recalculate quotas using MD and round down until the sum exactly matches the total seats.*

Example 2.35 From Table 2.29, apprion 30 seats to Palm Springs using Jefferson's Method.

Solution. The standard divisor (63) produced a sum of 29. For Jefferson's method, we must *decrease* the divisor to increase the quotas. Using a **Modified Divisor** of 61:

- A: $275/61 \approx 4.50 \to 4$
- B: $383/61 \approx 6.27 \to 6$
- C: $465/61 \approx 7.62 \to 7$
- D: $767/61 \approx 12.57 \to 13$

Sum: $4 + 6 + 7 + 13 = 30$. State D receives the extra seat.

Table 2.36 Jefferson's Method: Palm Springs

State	**A**	**B**	**C**	**D**	**Total**
Population	275	383	465	767	1890
Standard Quota (SD=63)	4.365	6.079	7.381	12.175	30
Lower Quota	4	6	7	12	29
Modified Quota (MD=61)	4.508	6.279	7.623	12.574	30.98
Modified LQ (Apportionment)	4	6	7	13	30

▲

Checkpoint 2.37 Apportioning 11 buses. Apportion 11 buses to the three towns using Jefferson's method.

Table 2.38 Bus Apportionment Table

Town	**Westboro**	**Shrewbury**	**Worcester**	**Total**
Population	89	120	168	377
Final Apportionment				

Solution. First, find the Standard Divisor: $SD = 377/11 \approx 34.27$.

Table 2.39 Trial 1: Using Standard Divisor (34.27)

Town	**Westboro**	**Shrewbury**	**Worcester**	**Total**
Population	89	120	168	377
Standard Quota	2.59	3.50	4.90	10.99
Lower Quota	2	3	4	9

The sum (9) is too low. We need 11 buses. We must *decrease* the divisor. Let's try a smaller divisor, $MD = 31$.

Table 2.40 Trial 2: Modified Divisor (31)

Town	**Westboro**	**Shrewbury**	**Worcester**	**Total**
Modified Quota	2.87	3.87	5.41	12.15
Lower Quota	2	3	5	10

The sum (10) is still too low. We must decrease the divisor further.

Trying $MD = 30$.

Table 2.41 Final Apportionment: Modified Divisor (30)

Town	**Westboro**	**Shrewbury**	**Worcester**	**Total**
Modified Quota	2.96	4.00	5.60	12.56
Modified LQ	2	4	5	11

The sum is exactly 11. Final Apportionment: Westboro: 2, Shrewbury: 4, Worcester: 5.

Table 2.42 Bus Apportionment Work Table

Town	**Westboro**	**Shrewbury**	**Worcester**	**Total**
Population	89	120	168	377
Standard Quota	2.59	3.50	4.90	10.99
Lower Quota	2	3	4	9
Modified Quota (MD=31)	2.87	3.87	5.41	12.15
Modified LQ	2	3	5	10
Modified Quota (MD=30)	2.96	4.00	5.60	12.56
Final Apportionment	2	4	5	11

Algorithm 2.43 Adam's Method.

1. *Find a* ***modified divisor****, d, so that when each group's* ***modified quota*** *is the upper quota, the sum of the whole numbers for all the groups is the number of items to be apportioned. The modified quotients that are rounded up are called* ***modified upper quota****.*

2. *Apportion to each group its modified upper quota.*

When using Adam's method, begin with a modified divisor that is slightly greater than the standard divisor.

Example 2.44 The Palm Springs is composed of four states: A, B, C, and D. According to the country's constitution, the congress will have 30 seats, divided among the four states according to their respective populations.

Find the standard quotas for states A, B, C, and D in Palm Springs and complete the table below. Use Adam's method with a modified divisor to apportion the congressmen.

Table 2.45 Population Data and Apportionment

State	**A**	**B**	**C**	**D**	**Total**
Population (1000s)	275	383	465	767	1890
Standard Quota					
Upper Quota					
Modified UQ					
Apportionment					30

Solution. 1. First, find the Standard Divisor (SD):

$$SD = \frac{1890}{30} = 63$$

2. Calculate Standard Quotas ($Pop/63$):

- A: 4.365 (Upper Quota: 5)
- B: 6.079 (Upper Quota: 7)

- C: 7.381 (Upper Quota: 8)
- D: 12.175 (Upper Quota: 13)

Sum of Upper Quotas $= 5+7+8+13 = 33$ (Too many seats).

3. Since the sum is too high, we increase the divisor. Let's try $md = 70.5$:

- A: $275/70.5 \approx 3.90 \to 4$
- B: $383/70.5 \approx 5.43 \to 6$
- C: $465/70.5 \approx 6.60 \to 7$
- D: $767/70.5 \approx 10.88 \to 11$

Sum $= 4+6+7+11 = 28$ (Too low).

4. Refined $md = 67.5$: Sums will result in A:5, B:6, C:7, D:12 $= 30$.
Using $md = 66.5$:

State	**A**	**B**	**C**	**D**	**Total**
Standard Quota ($d = 63$)	4.37	6.08	7.38	12.18	30
Upper Quota	5	7	8	13	33
Mod. Quota ($d = 66.5$)	4.14	5.76	6.99	11.53	—
Modified UQ	5	6	7	12	30

▲

Finding the Modified Divisor for Adam's Method:.

1. Compute the standard divisor

$$D = \frac{\text{Total population}}{\text{Number of seats}}.$$

2. Increase D by an amount so that when allocations (State population / Modified D) are rounded upward, they add up to the exact number of seats.

Checkpoint 2.46 Apportioning 7 Doctors. Apportion 7 doctors to 4 clinics using Adam's method.

Clinic	**West**	**North**	**South**	**East**	**Total**
Population	120	180	160	240	700
Appointment					

Solution. Standard Divisor $(SD) = 700/7 = 100$.

The final apportionment using Adam's method with a modified divisor of md=120 is: West: 1, North: 2, South: 2, and East: 2.

Clinic	**West**	**North**	**South**	**East**	**Total**
Population	120	180	160	240	700
Standard Quota ($d = 100$)	1.2	1.8	1.6	2.4	7
Upper Quota (Initial)	2	2	2	3	9 (Too High)
Mod. Quota ($d = 120$)	1.0	1.5	1.33	2.0	—
Modified UQ (Final)	1	2	2	2	7

Checkpoint 2.47 Apportioning 40 HMO Doctors. An HMO has 40 doctors to be apportioned among four clinics: A, B, C, and D.

Clinic	**A**	**B**	**C**	**D**
Patient Load	275	392	611	724
Apportionment				

1 Use Hamilton's method.

2 Use Jefferson's method with $d = 48$.

3 Use Adam's method.

Solution 1 (Hamilton's Method Solution). Calculate Standard Divisor (SD) and Quotas

- Total Patient Load: $275 + 392 + 611 + 724 = 2002$
- $SD = 2002/40 = 50.05$

Clinic	**A**	**B**	**C**	**D**	**Total**
Load	275	392	611	724	2002
Standard Quota	5.4945	7.8322	12.2078	14.4655	40
Lower Quota	5	7	12	14	38
Fractional Part	0.4945	0.8322	0.2078	0.4655	—
Surplus Seat	+1	+1	0	0	+2
Apportionment	6	8	12	14	40

Solution 2 (Jefferson's Method Solution (d=48)).

Clinic	**A**	**B**	**C**	**D**	**Total**
Modified Quota	5.73	8.17	12.73	15.08	—
Apportionment	5	8	12	15	40

Solution 3 (Adam's Method Solution (md=52)).

Clinic	**A**	**B**	**C**	**D**	**Total**
Modified Quota	5.29	7.54	11.75	13.92	—
Apportionment	6	8	12	14	40

Checkpoint 2.48 Apportioning Math Course Sections. Thirty sections of math courses are to be offered in introductory algebra, intermediate algebra, college algebra, and liberal arts math. The preregistration figures for the number of students planning to enroll in each section are given in the following table.

Course	**Intro**	**Intermed**	**College**	**Lib Arts**	**Total**
Enrollment	388	405	213	345	1351

1 Use Hamilton's method to determine how many sections of each course should be offered.

2 Use Jefferson's method with Modified divisor md = _____ to determine how many sections of each course should be offered.

3 Use Adam's method with Modified divisor md = _____ to determine how many sections of each course should be offered.

Solution. Thirty sections of math courses are to be offered based on preregistration figures.

1 Hamilton's Method (SD = 45.0333):

Course	**Intro**	**Intermed**	**College**	**Lib Arts**	**Total**
Enrollment	388	405	213	345	1351
Standard Quota	8.6158	8.9933	4.7298	7.6610	30
Lower Quota	8	8	4	7	27
Fractional Part	0.6158	0.9933	0.7298	0.6610	—
Surplus Row	0	+1	+1	+1	+3
Apportionment	8	9	5	8	30

2 Jefferson's Method (using $d = 42.5$):

Course	**Intro**	**Intermed**	**College**	**Lib Arts**	**Total**
Mod. Quota	9.129	9.529	5.011	8.117	—
Apportionment	9	9	5	8	31 (Try Again)
Mod. Quota ($d = 43$)	9.023	9.418	4.953	8.023	—
Apportionment	9	9	4	8	30

3 Adam's Method (using $d = 48$):

Course	**Intro**	**Intermed**	**College**	**Lib Arts**	**Total**
Mod. Quota	8.083	8.437	4.437	7.187	—
Apportionment	9	9	5	8	31 (Try Again)
Mod. Quota ($d = 49$)	7.918	8.265	4.346	7.040	—
Apportionment	8	9	5	8	30

Example 2.49 Hamilton's Method Apportionment of 100 seats. Suppose there were four states, A, B, C ,and D with populations as shown in the table below. Suppose these four states decide to create a legislature with 100 seats. Apply Hamilton's method to determine the apportionment.

Table 2.50

Category	**State A**	**State B**	**State C**	**State D**	**Total**
Population	936	2,726	2,603	3,735	10,000
Standard Quota					
Lower Quota					
Surplus					
Apportioned Seats					100

The following table demonstrates the apportionment using Hamilton's Method with a standard divisor of $SD = 10,000/100 = 100$.

Table 2.51 Hamilton's Method Apportionment

Category	State A	State B	State C	State D	Total
Population	936	2,726	2,603	3,735	10,000
Standard Quota	9.36	27.26	26.03	37.35	100.00
Lower Quota	9	27	26	37	99
Fractional Part	0.36	0.26	0.03	0.35	-
Surplus	+1				
Final Seats	10	27	26	37	100

▲

Theorem 2.52 The Alabama Paradox. *In a fair apportionment system, adding extra seats must not result in fewer seats for any state. The **Alabama Paradox** occurs when the total number of available seats increases, yet one state (or more) loses seats as a result. This paradox can occur when using* HAMILTON'S METHOD.

Example 2.53 Hamilton's Method Apportionment of 101 seats. Using Hamilton's method, recompute the apportionment from Example 2.49, if there are 101 seats instead of 100. Does the Alabama paradox occur?

Table 2.54

States	State A	State B	State C	State D	Total
Population	936	2,726	2,603	3,735	10,000
Standard Quota					
Lower Quota					
Surplus					
Apportioned Seats					101

The table below shows the Hamilton's Method apportionment with 101 seats.

Note that State A loses a seat compared to the 100-seat scenario, demonstrating the Alabama Paradox.

Table 2.55 Hamilton's Method (101 Seats)

States	State A	State B	State C	State D	Total
Population	936	2,726	2,603	3,735	10,000
Standard Quota	9.4536	27.5326	26.2903	37.7235	101.0000
Lower Quota	9	27	26	37	99
Fractional Part	0.4536	0.5326	0.2903	0.7235	-
Surplus		+1		+1	
Final Seats	9	28	26	38	101

▲

Chapter Exercise

Exercises

1. The city council gave a questionnaire to its citizens, asking them their priorities for next year's budget: (P)olice, (R)oads, (S)chools, and (T)rash removal.

Preference	**15**	**30**	**18**	**17**	**2**
1st	P	T	P	R	R
2nd	R	R	S	T	S
3rd	S	S	T	S	P
4th	T	P	R	P	T

(a) Which option is selected using the plurality method?

(b) Which option is selected using the plurality-with-elimination method?

(c) Which option is selected using the Borda count method?

(d) Which option is selected using the pairwise comparison method?

Solution.

(a) Plurality Method: Count only first-place votes.

- P: $15 + 18 = 33$
- T: 30
- R: $17 + 2 = 19$
- S: 0

Winner: Police (P) with 33 votes.

(b) Plurality-with-Elimination:

- Round 1: S has 0 votes. Eliminate S.

Preference	**15**	**30**	**18**	**17**	**2**
1st	P	T	P	R	R
2nd	R	R	T	T	P
3rd	T	P	R	P	T

- Round 2: R has the fewest 1st place votes (19). Eliminate R. The 17 votes from (R, T, S, P) move to T. The 2 votes from (R, S, P, T) move to P.

Preference	**15**	**30**	**18**	**17**	**2**
1st	P	T	P	T	P
2nd	T	P	T	P	T

New totals: P: $15 + 18 + 2 = 35$; T: $30 + 17 = 47$.

Winner: Trash Removal (T).

(c) Borda Count: 1st = 4pts, 2nd = 3pts, 3rd = 2pts, 4th = 1pt.

- P: $15(4) + 30(1) + 18(4) + 17(1) + 2(2) = 183$
- R: $15(3) + 30(3) + 18(1) + 17(4) + 2(4) = 229$
- S: $15(2) + 30(2) + 18(3) + 17(2) + 2(3) = 184$
- T: $15(1) + 30(4) + 18(2) + 17(3) + 2(1) = 224$

Winner: Roads (R) with 229 points.

(d) Pairwise Comparison:

- P vs R: P=33, R=49. (R wins)

- P vs S: P=63, S=19. (P wins)
- P vs T: P=35, T=47. (T wins)
- R vs S: R=64, S=18. (R wins)
- R vs T: R=52, T=30. (R wins)
- S vs T: S=35, T=47. (T wins)

R has 3 wins, T has 2 wins, P has 1 win. Winner: Roads (R).

2. The preference table shows the results of an election among three candidates, A, B, and C.

Number of votes	**7**	**6**	**3**
1st choice	A	B	C
2nd choice	B	C	B
3rd choice	C	A	A

(a) Using the plurality method, who is the winner?

(b) Is the Majority satisfied?

(c) Is the Condorcet satisfied?

Solution. Total votes: $7 + 6 + 3 = 16$. Votes needed for a majority: $16/2 = 8$, so 9 votes are required.

(a) Plurality Method: Count only 1st choice votes:

- A: 7 votes
- B: 6 votes
- C: 3 votes

The winner is Candidate A.

(b) Majority Criterion: A candidate must have more than 50% of the first-place votes to be a majority winner. Since the leader (A) has only 7 votes and 9 were needed for a majority, there is no majority winner. The criterion is satisfied (vacuously) because the criterion only says "if a majority winner exists, they must win." Since none exists, the rule isn't broken.

(c) Condorcet Criterion: We check if the plurality winner (A) is also the Condorcet winner by doing pairwise comparisons:

- A vs B: A is preferred by 7 voters; B is preferred by $6 + 3 = 9$ voters. B wins.
- B vs C: B is preferred by $7 + 3 = 10$ voters; C is preferred by 6 voters. B wins.
- A vs C: A is preferred by 7 voters; C is preferred by $6 + 3 = 9$ voters. C wins.

Candidate B wins all head-to-head matches and is the Condorcet Winner. Since the plurality method chose A instead of B, the Condorcet Criterion is not satisfied in this election.

3. The preference table shows the results of an election among three candidates, A, B, and C.

Number of votes	**10**	**4**	**2**
1st choice	A	B	C
2nd choice	B	C	B
3rd choice	C	A	A

(a) Using the plurality-with-elimination method, who is the winner?

(b) Is the Majority satisfied?

(c) In the actual election, the 2 voters in the last column who voted C, B, and A, change their vote to A, B, C. Using the plurality-with-elimination method, which candidate wins the actual election?

Solution. Total votes: $10 + 4 + 2 = 16$. Majority needed: 9 votes.

(a) Plurality-with-Elimination:

- Round 1: A has 10 votes, B has 4, and C has 2. Since A has 10 votes (more than the majority of 9), A is the winner immediately.

Winner: Candidate A.

(b) Majority: Yes. Candidate A has a majority of first-place votes (10 out of 16) and won the election. Therefore, the Majority is satisfied.

(c) Changed Election: If the 2 voters for C change to A, the new totals are:

- A: $10 + 2 = 12$ votes
- B: 4 votes
- C: 0 votes

Winner: Candidate A.

4. If a candidate wins all pairwise comparisons with other candidates, that candidate is called the ***Condorcet winner***.

The mathematics department is hiring a new faculty member and the five-person hiring committee has interviewed four candidates: Adam(A), Beth(B), Carol(C), and Dan(D). They have decided to use Condorcet's method on their five ballots. Who gets the offer?

Choice	**Voter 1**	**Voter 2**	**Voter 3**	**Voter 4**	**Voter 5**
First	B	D	C	D	B
Second	A	B	A	A	A
Third	C	C	B	C	D
Fourth	D	A	D	B	C

Solution. To find the Condorcet winner, we must compare each candidate head-to-head. There are 5 voters, so 3 votes are needed to win a comparison.

- A vs. B:

  Voter 1: B, Voter 2: B, Voter 3: A, Voter 4: A, Voter 5: B.

  Result: B wins 3 to 2.

- A vs. C:

  Voter 1: A, Voter 2: C, Voter 3: C, Voter 4: A, Voter 5: A.

Result: A wins 3 to 2.

- A vs. D:
  Voter 1: A, Voter 2: D, Voter 3: A, Voter 4: D, Voter 5: A.
  Result: A wins 3 to 2.
- B vs. C:
  Voter 1: B, Voter 2: B, Voter 3: C, Voter 4: C, Voter 5: B.
  Result: B wins 3 to 2.
- B vs. D:
  Voter 1: B, Voter 2: D, Voter 3: B, Voter 4: D, Voter 5: B.
  Result: B wins 3 to 2.
- C vs. D:
  Voter 1: C, Voter 2: D, Voter 3: C, Voter 4: D, Voter 5: D.
  Result: D wins 3 to 2.

Summary of Wins:

- Beth (B) won 3 comparisons (vs. A, C, and D).
- Adam (A) won 2 comparisons (vs. C and D).
- Dan (D) won 1 comparison (vs. C).
- Carol (C) won 0 comparisons.

Since Beth (B) won every head-to-head match, she is the Condorcet winner and gets the offer.

5. A small city has 30 police officers to be apportioned among 4 precincts based on the population of each precinct. The populations are given in the following table.

Precinct	**1**	**2**	**3**	**4**	**Total**
Population	3117	1883	2776	3174	10,950

(a) Find the standard divisor.

(b) Find the standard quota for each precinct.

(c) Find the modified quota for the fourth precinct using Jefferson's method with a modified divisor of 350.

(d) Find the apportionment for the second precinct using Adams's method with a modified divisor of 380.

Solution. Part (a) & (b): Standard Divisor and Standard Quotas
The Standard Divisor (SD) is $10,950/30 = 365$.

Table 2.56 Hamilton's Method (Standard Quotas)

Category	Precinct 1	Precinct 2	Precinct 3	Precinct 4	Total
Population	3117	1883	2776	3174	10,950
Std. Quota	8.54	5.16	7.61	8.70	30.00
Lower Quota	8	5	7	8	28
Fraction	0.54	0.16	0.61	0.70	-
Surplus			+1	+1	
Hamilton Final	8	5	8	9	30

Part (c): Jefferson's Method (Modified Divisor $d = 350$)

Table 2.57 Jefferson's Method Calculation

Category	Precinct 1	Precinct 2	Precinct 3	Precinct 4	Total
Mod. Quota	8.91	5.38	7.93	9.07	-
Jefferson Apport.	8	5	7	9	29

The modified quota for the fourth precinct is 9.07.
Part (d): Adams's Method (Modified Divisor $d = 380$)

Table 2.58 Adams's Method Calculation

Category	Precinct 1	Precinct 2	Precinct 3	Precinct 4	Total
Mod. Quota	8.20	4.96	7.31	8.35	-
Adams Apport.	9	5	8	9	31

The apportionment for the second precinct using Adams's method is 5 officers.

6. A mother has an incentive program to get her five children to read more. She has 30 pieces of candy to divide among her children at the end of the week based on the number of minutes each of them spends reading.

Child	Abby	Bobby	Charli	Dave	Eddie	Total
Reading Times	138	142	188	218	64	750

(a) Use Hamilton's method to apportion 30 pieces of candy.

(b) At the last minute, the mother finds another piece of candy and does the apportionment again. Now, she has 31 pieces of candy to divide among her children at the end of the week based on the number of minutes each of them spends reading. Use Hamilton's method to apportion the candy among the children.

(c) Look at your answers for parts (a) and (b). What changed and why? Why is the change interesting?This is an example of which paradox?

Solution. Part (a): 30 Pieces (Standard Divisor = 25)

Table 2.59 Apportionment for 30 Pieces

Category	Abby	Bobby	Charli	Dave	Eddie	Total
Reading Min	138	142	188	218	64	750
Std. Quota	5.52	5.68	7.52	8.72	2.56	30.00
Lower Quota	5	5	7	8	2	27
Fraction	0.52	0.68	0.52	0.72	0.56	-
Surplus		+1		+1	+1	
Final Seats	5	6	7	9	3	30

Part (b): 31 Pieces (Standard Divisor = 24.1935)

Table 2.60 Apportionment for 31 Pieces

Category	**Abby**	**Bobby**	**Charli**	**Dave**	**Eddie**	**Total**
Reading Min	138	142	188	218	64	750
Std. Quota	5.704	5.870	7.771	9.011	2.645	31.00
Lower Quota	5	5	7	9	2	28
Fraction	0.704	0.870	0.771	0.011	0.645	-
Surplus	+1	+1	+1			
Final Seats	6	6	8	9	2	31

Part (c): Increasing the total number of items from 30 to 31 creates the **Alabama Paradox**. In this specific case, Eddie loses a candy even though the total amount of candy increased.

Chapter 3

Personal Finance

In an era of digital banking and shifting economies, understanding your personal finances is a superpower. This resource moves beyond the classroom to show how smart financial habits today can create long-term stability and open doors for the opportunities you care about most.

Understanding Interest: Simple Interest

Interest is the dollar amount that we get paid for lending money or pay for borrowing money. When we deposit money in a financial institution, the institution pays us interest for its use. When we borrow money, interest is the price we pay for the privilege of using the money until we repay it.

Definition 3.1 Simple Interest. The amount of money that we deposit or borrow is called the ***principal***. The amount of interest depends on the principal, the interest ***rate***, which is given as a percent and various from bank to bank, and the length of time for which the money is deposited. ***Simple interest*** involves interest calculated ***only on the principal***.

$$Interest = Principal \times rate \times time$$

$$I = Prt$$

The rate, r, is expressed as a decimal when you calculating simple interest. ◆

Example 3.2 Calculating Simple interest for a year. You deposit \$2,000 in a savings account, which has a rate 6.0%. Find the interest at the end of the first year.We can use the formula for simple interest to find the interest at the end of the first year. $Interest = Principal \times rate \times time$ $Interest = 2000 \times 0.06 \times 1 = 120$ ▲

Example 3.3 Calculating Simple interest for more than a year. A student took out a simple interest loan for \$1,300 for two years at a rate of 7.5%to purchase a used car. What is the interest on the loan?We can use the formula for simple interest to find the interest on the loan. $Interest = Principal \times rate \times time$ $Interest = 1300 \times 0.075 \times 2 = 195$ ▲

When a loan is repaid, the interest is added to the original principal to find the total amount due. In Example 3.3, at the end of 2 years the student will have to repay.

$Principal + Interest = 1300 + (1300 \times 0.075 \times 2) = 1300 + 195 = 1495$

In general, after t years the amount due, A, can be determined as follows:

$$A = P + \underbrace{I}_{I=Prt} = P + Prt = P(1+rt)$$

Definition 3.4 Future Value. ***Future value = Principal + Interest***: The amount due, A, is called the ***future value*** of the loan. The principal borrowed P is also known as the loan's ***present value***.

$$A = P(1+rt)$$

◆

Example 3.5 Calculating Future Value. What is the future value of a $1,000 loan at 5% for 3 years?We can use the formula for future value to find the amount due at the end of 3 years. $A = P(1+rt)$ $A = 1000(1+0.05\times 3) = 1000(1+0.15) = 1000\times 1.15 = 1150$ ▲

Example 3.6 Determining a simple interest rate. You borrow $2,500 from a friend and promise to pay back $2,655 in six months. What simple interest rate will you pay?We can use the formula for future value to find the interest rate.

$$\begin{aligned} A &= P(1+rt) \\ 2655 &= 2500(1+r\times 0.5) \\ 2655 &= 2500 + 1250r \\ 1250r &= 155 \\ r &= \frac{155}{1250} \\ r &= 0.124 = 12.4\% \end{aligned}$$

▲

Example 3.7 Determining a present value. How much should you put an investment paying a simple interest rate of 8.0% if you need $3000 in six months?We can use the formula for future value to find the present value.

$$\begin{aligned} A &= P(1+rt) \\ 3000 &= P(1+0.08\times 0.5) \\ 3000 &= P(1+0.04) && = P(1.04) \\ P &= \frac{3000}{1.04} \\ P &= 2884.62 \end{aligned}$$

▲

Exercises for Practice

1. **Calculating a simple interest rate for a T-bill.** Treasury bills (T-bills) can be purchased from the U.S. Treasury Department. You buy a T-bill for $981.60 that pays $1,000 in 13 weeks. What simple interest rate, to the nearest tenth of a percent, does this T-bill earn?

   Answer. We can use the formula for future value to find the interest rate.

$$A = P(1+rt)$$

$$\begin{aligned}
1000 &= 981.60(1 + r \times \frac{13}{52}) \\
1000 &= 981.60(1 + 0.25r) \\
1000 &= 981.60 + 245.40r \\
245.40r &= 18.40 \\
r &= \frac{18.40}{245.40} = 0.0749 = 7.5\%
\end{aligned}$$

2. **Calculating a present value for a CD.** A bank offers a CD that pays a simple interest rate of 4.5%. How much must you put in this CD now in order to have \$8,000 for a kitchen remodeling project in two years? Round up to the nearest cent.

   Answer. We can use the formula for future value to find the present value.

$$\begin{aligned}
A &= P(1 + rt) \\
8000 &= P(1 + 0.045 \times 2) \\
8000 &= P(1 + 0.09) = P(1.09) \\
P &= \frac{8000}{1.09} = 7339.45
\end{aligned}$$

3. **Calculating a future value for a savings account.** You deposit \$1,000 in a savings account at a bank that has a rate of 1.95%. Find a future value at the end of 5 years.

   Answer. We can use the formula for future value to find the amount due at the end of 5 years. $A = P(1 + rt)$ $A = 1000(1 + 0.0195 \times 5) = 1000(1 + 0.0975) = 1000 \times 1.0975 = 1097.50$

4. **Determining whether a statement makes sense or not, explain your reasoning..**

   (a) After depositing \$1,500 in an account at a rate of 4%, my balance at the end of the first year was \$(1500)(0.04).

   (b) I planned to save \$5,000 in four years, computed the present value to be \$3,846.153, so I round the principal to \$3,846.15.

   Answer.

   (a) This statement does not make sense because the balance at the end of the first year should be the future value, which is the sum of the principal and interest. The correct balance should be $\$1500 + (1500)(0.04) = \$1560$.

   (b) This makes sense. In fact, rounding down is the safer move here. Rounding Down (\$3,846.15): By starting with slightly less than the exact "perfect" number, you technically might end up a fraction of a cent short of your \$5,000 goal after four years. However, in practice, this is unlikely to be a significant issue, and it provides a small buffer to ensure you meet your target.

Saving Money: Compound Interest

In 1626, Peter Minuit convinced the wrappinger Indians to sell him Manhattan Island for \$24. If the native American had ***put the \$24 into a bank account***

at a 5% interest rate compound annually, by the year 2021 there would be well over \$5.6 billion in the account!

So, how did the present value of Manhattan in 1626 attain a future value of over \$5.6 billion in 2021, 395 years at 5% simple interest is

$$A = P(1 + rt) = 24(1 + (0.05)(395)) = 498.00$$

or \$498.00 compared to over \$5.6 billion.

Compound Interest vs. Simple Interest.

The difference between simple interest and compound interest is that simple interest is calculated only on the original principal, while compound interest is calculated on the original principal and any accumulated interest.

Compound interest is interest computed on the original principal as well as on any accumulated interest.

For example, suppose you deposit \$1,000 in a savings account at a rate of 5%. Following table shows how the investment grows.

Calculating Compound interest for a year.

Table 3.8 Compound Interest Growth of \$1000 in a savings account at a rate of 5%

Year	Beginning Balance (\$)	Interest Earned: $I = Prt$	New Balance(\$)
1	\$1,000.00	$I = \$1,000.00 \times 0.05 \times 1 = 50$	\$1,050.00
2	\$1,050.00	$I = \$1,050.00 \times 0.05 \times 1 = 52.50$	\$1,102.50
3	\$1,102.50	$I = \$1,102.50 \times 0.05 \times 1 = 55.13$	\$1,157.63
4	\$1,157.63	$I = \$1,157.63 \times 0.05 \times 1 = 57.88$	\$1,215.51
5	\$1,215.51		
6			

The fast way to determine the amount, A, in the account subject to compound interest is the following formula. If P dollars are deposited at rate r, in decimal form, subject to compound interest, then the amount, A, of money in the account after t years is given by

Definition 3.9 Compound Interest Paid Once a Year. Calculating the amount in an account ***for compound interest paid once year:***

$$A = P(1 + r)^t$$

where the amount A is the account's ***future value*** and the principal P is its ***present value***. ◆

Example 3.10 Calculating Compound interest for a year. You deposit \$2,000 in a savings account, which has a rate 6%. Find the amount, A, of money in the account after 3 years subject to compound interest and find the interest.To find the amount A after 3 years, we use the compound interest formula

$$A = P(1 + r)^t$$

where $P = 2000$, $r = 0.06$, and $t = 3$.

Substituting the values:

$$A = 2000(1 + 0.06)^3 = 2000(1.06)^3 = \$2382.03$$

The interest earned is calculated as:

$$I = A - P = 2382.03 - 2000 = \$382.03$$

▲

Calculating the amount in an account ***for compound interest paid more than once a year.*** The period of time between two interest payments is called the ***compounding period***.

If compound interest is paid once a year, the compounding period is one year. We say the interest is ***compounded annually***.

If compound interest is paid twice a year, the compounding period is six months. We say the interest is ***compounded semiannually***.

If compound interest is paid four times a year, the compounding period is three months. We say the interest is ***compounded quarterly***.

If compound interest is paid every month of a year, the compounding period is twelve months. We say the interest is ***compounded monthly***.

In general, if compound interest is paid $\boldsymbol{n}$ times per year, we say the interest is ***n compounding periods per year***.

Definition 3.11 Compound interest: Future Value. If P dollars are deposited at rate r, in decimal form, subject to compound interest paid n times per year, then the amount, A, of money in the account after t years is given by

$$A = P\left(1 + \frac{r}{n}\right)^{nt} \tag{3.1}$$

where the amount A is the account's ***future value*** and the principal P is its ***present value***. ◆

Example 3.12 Calculating Compound Interest for More Than a Year. You deposit \$7,500 in a savings account that has a rate of 6%. The interest is compounded monthly. How much money will you have after five years? Find the interest after five years.

Solution. To find the future value of the account, we use the compound interest formula:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

Given $P = 7500$, $r = 0.06$, $n = 12$, and $t = 5$:

$$\begin{aligned} A &= 7500\left(1 + \frac{0.06}{12}\right)^{12\times5} \\ &= 7500(1.005)^{60} \\ &\approx 10116.38 \end{aligned}$$

The account balance after five years is approximately \$10,116.38. The interest earned over five years is:

$$I = 10116.38 - 7500 = 2616.38$$

The interest earned is \$2,616.38. ▲

Definition 3.13 Compound Interest: Present Value. If A dollars are to be accumulated in t years in an account that pays rate r compounded n times per year, then the present value, P, that needs to be invested now is given by

$$P = \frac{A}{\left(1 + \frac{r}{n}\right)^{nt}}$$

◆

Example 3.14 Calculating present value. How much money should be deposited in an account today that earns 8% compounded monthly so that it will accumulate to \$20,000 in five years?To find the required deposit (Present Value), we use the formula:

$$P = \frac{A}{(1+\frac{r}{n})^{nt}}$$

where:

- $A = 20,000$ (Future Value)
- $r = 0.08$ (Annual interest rate)
- $n = 12$ (Compounding periods per year)
- $t = 5$ (Time in years)

Substituting these values into the formula gives:

$$P = \frac{20,000}{(1+\frac{0.08}{12})^{12\times 5}}$$
$$P = \frac{20,000}{(1.0066667)^{60}}$$
$$P \approx 13,424.21$$

Therefore, a deposit of $\$13,424.21$ is required today. ▲

The ***annual percentage yield (APY),*** or the ***effective interest rate***, is the simple interest rate that produces the same amount of money in an account at the end of one year as when the account is subject to compound interest at a stated rate.

Definition 3.15 Annual percentage yield (APY). Suppose that an investment has a nominal interest rate, r, in decimal form, and pays compound interest n times per year. The investment's ***effective annual yields***, APY, is given by

$$Y = \left(1+\frac{r}{n}\right)^n - 1 \tag{3.2}$$

◆

Example 3.16 Understanding effective annual yield. You deposit \$4000 in an account that pays 8% interest compounded monthly. Use the future value formula for simple interest to determine the effective annual yield.By the formula for future value

$$A = P(1+\frac{r}{n})^{nt} = 4000(1+\frac{0.08}{12})^{12\cdot 1} \approx \$4,332.00$$

Rounded to nearest cent, the amount in the savings account after one year in \$4332.00.

The effective annual yield is a simple interest rate. We use the future value formula for simple interest to determine the simple interest rate that produces a future value of \$4332 for \$4000 deposit after one year.

$$A = P(1+rt)$$
$$4332 = 4000(1+r(1))$$
$$4332 = 4000 + 4000r$$

$$332 = 4000r$$
$$r = 0.083 = 8.3\%$$

You will pay a simple interest rate of 8.3%. This means that money invested at 8.3% simple interest earns the same amount in one year as money invested at 8%interest compounded monthly. ▲

Example 3.17 Calculating Effective Annual Yield. A passbook savings account has a nominal rate of 5%. The interest is compounded daily. Find the account's effective annual yield, assuming a 365-day year.

Solution. The Effective Yield (APY) represents the actual interest earned in one year. It is calculated using the formula:

$$Y = \left(1 + \frac{r}{n}\right)^n - 1$$

where:

- $r = 0.05$ (Nominal annual interest rate)
- $n = 365$ (Compounding periods per year)

Substituting the given values:

$$Y = \left(1 + \frac{0.05}{365}\right)^{365} - 1$$
$$Y \approx (1.000136986)^{365} - 1$$
$$Y \approx 1.051267 - 1$$
$$Y \approx 0.051267$$

Converted to a percentage, the effective annual yield is approximately 5.13%. ▲

Example 3.18 Quarterly Compounding and Interest Calculation. You deposit $\$1,000$ in a savings account at a bank that has a rate of 2.95%. The interest is compounded quarterly.

a How much money will you have after ten years? Round up to the nearest cent.

b Find the interest.

Solution. First, we identify the variables for the compound interest formula:

- $P = 1,000$ (Principal)
- $r = 0.0295$ (Annual rate)
- $n = 4$ (Quarterly compounding)
- $t = 10$ (Years)

For part (a), we calculate the future value (A):

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$
$$A = 1,000\left(1 + \frac{0.0295}{4}\right)^{4\times 10}$$
$$A = 1,000(1.007375)^{40}$$

$$A \approx 1,341.517$$

Rounding up, the account balance will be $1,341.52.

For part (b), we find the interest earned (I) by subtracting the principal from the future value:

$$\begin{aligned} I &= A - P \\ I &= 1,341.52 - 1,000 \\ I &= 341.52 \end{aligned}$$

The interest earned is $341.52. ▲

Checkpoint 3.19 Comparing Investment Options. Suppose that you have $10,000 to invest. Which investment yields the greater return over 5 years: 7.55% compounded quarterly or 7.6% compounded semiannually?

Solution. To determine which investment is superior, we calculate the future value (A) for both options using the formula $A = P(1 + \frac{r}{n})^{nt}$, where $P = 10,000$ and $t = 5$.

Option 1: 7.55% **compounded quarterly** Here, $r = 0.0755$ and $n = 4$.

$$\begin{aligned} A &= 10,000\left(1 + \frac{0.0755}{4}\right)^{4\times5} \\ A &= 10,000(1.018875)^{20} \\ A &\approx 14,530.01 \end{aligned}$$

Option 2: 7.6% **compounded semiannually** Here, $r = 0.076$ and $n = 2$.

$$\begin{aligned} A &= 10,000\left(1 + \frac{0.076}{2}\right)^{2\times5} \\ A &= 10,000(1.038)^{10} \\ A &\approx 14,520.23 \end{aligned}$$

Comparing the two totals, Option 1 results in $14,530.01 while Option 2 results in $14,520.23.

Therefore, the 7.55% investment compounded quarterly yields the greater return by $9.78.

Checkpoint 3.20 Present Value with Quarterly Compounding. How much money should be deposited today in an account that earns 5% compounded quarterly so that it will accumulate to $15,000 in ten years?

Solution. To find the required initial deposit, we use the Present Value formula:

$$P = \frac{A}{(1 + \frac{r}{n})^{nt}}$$

where:

- $A = 15,000$ (The target future value)
- $r = 0.05$ (The annual interest rate)
- $n = 4$ (Compounded quarterly)
- $t = 10$ (The time in years)

Substituting these values into the formula:

$$\begin{aligned}
P &= \frac{15,000}{(1+\frac{0.05}{4})^{4\times 10}} \\
P &= \frac{15,000}{(1.0125)^{40}} \\
P &\approx \frac{15,000}{1.643619} \\
P &\approx 9,126.20
\end{aligned}$$

Therefore, a deposit of $\$9,126.20$ is required today to reach the goal of $\$15,000$ in ten years.

Exercises for Practice

1. **Calculating Effective Annual Yield.** You deposit $\$7,500$ in an account that pays 4.75% interest compounded monthly. Determine the effective annual yield.

Solution. The Effective Annual Yield (APY) is the actual annual rate of return and is independent of the principal amount deposited. We use the formula:

$$Y = \left(1+\frac{r}{n}\right)^n - 1$$

where:

- $r = 0.0475$ (Nominal annual interest rate)
- $n = 12$ (Compounding periods per year)

Substituting these values:

$$\begin{aligned}
Y &= \left(1+\frac{0.0475}{12}\right)^{12} - 1 \\
Y &\approx (1.0039583)^{12} - 1 \\
Y &\approx 1.048545 - 1 \\
Y &\approx 0.048545
\end{aligned}$$

Converted to a percentage and rounded to two decimal places, the effective annual yield is 4.85%.

2. **Future Value for a Long-Term Investment.** At the time of her grandson's birth, a grandmother deposits $\$5,000$ in an account that pays 6% compounded monthly. What will be the value of the account at the child's twenty-first birthday, assuming that no other deposits or withdrawals are made during this period?

Solution. To find the future value of the account, we use the compound interest formula:

$$A = P\left(1+\frac{r}{n}\right)^{nt}$$

where:

- $P = 5,000$ (Initial deposit)
- $r = 0.06$ (Annual interest rate)
- $n = 12$ (Compounding periods per year)

- $t = 21$ (Time in years until the 21st birthday)

Substituting these values into the formula:

$$A = 5,000\left(1 + \frac{0.06}{12}\right)^{12\times21}$$
$$A = 5,000(1.005)^{252}$$
$$A \approx 5,000(3.514418)$$
$$A \approx \$17,572.09$$

On the child's twenty-first birthday, the account will be worth $\$17,572.09$.

3. Interest Rate Concepts: True or False. Determine whether each statement is True or False and explain your reasoning.

1 My bank provides simple interest at 3.25% per year, but I cannot determine if this is a better deal than a competing bank offering 3.25% compound interest without knowing the compounding period.

2 My bank advertises a compound interest rate of 2.4%, although, without making deposits or withdrawals, the balance in my account increased by 2.43% in one year.

Solution.

1 False. Any amount of compounding is better than simple interest when the nominal rates are identical (and greater than zero). While the compounding period determines *how much* better the deal is, we already know compound interest will yield more than simple interest regardless of whether it is compounded annually, monthly, or daily.

2 True. The advertised rate is the nominal rate (2.4%), but the 2.43% increase represents the *Effective Annual Yield*. Because interest is added to the balance throughout the year, the actual growth is slightly higher than the nominal rate. For example, 2.4% compounded monthly results in an effective yield of approximately 2.427%.

4. Quarterly Compounding vs. Simple Interest. A sum of $\$10,000$ is invested at an annual rate of 3%. Find the balance in the account after 5 years subject to:

1 quarterly compounding

2 simple interest

Solution. We are given $P = 10,000$, $r = 0.03$, and $t = 5$.

1 *Quarterly Compounding.*
Using the formula $A = P(1 + \frac{r}{n})^{nt}$ with $n = 4$:

$$A = 10,000\left(1 + \frac{0.03}{4}\right)^{4\times5}$$
$$A = 10,000(1.0075)^{20}$$
$$A \approx 10,000(1.161184)$$
$$A \approx 11,611.84$$

The balance with quarterly compounding is $\$11,611.84$.

2 *Simple Interest.*

Using the formula $A = P(1 + rt)$:

$$\begin{aligned}
A &= 10,000(1 + 0.03 \times 5) \\
A &= 10,000(1 + 0.15) \\
A &= 10,000(1.15) \\
A &= 11,500.00
\end{aligned}$$

The balance with simple interest is $\$11,500.00$.

Quarterly compounding results in an additional $\$111.84$ compared to simple interest.

5. Future Value and Effective Yield. You deposit $\$3,000$ in an account that pays 4.5% interest compounded monthly.

1 Find the future value after one year.

2 Determine the effective annual yield.

Solution. Given: $P = 3,000$, $r = 0.045$, $n = 12$, and $t = 1$.

1 To find the future value (A) after one year:

$$\begin{aligned}
A &= P\left(1 + \frac{r}{n}\right)^{nt} \\
A &= 3,000\left(1 + \frac{0.045}{12}\right)^{12\times 1} \\
A &= 3,000(1.00375)^{12} \\
A &\approx 3,000(1.0459398) \\
A &\approx 3,137.82
\end{aligned}$$

The future value after one year is $\$3,137.82$.

2 To find the effective annual yield (EAY):

$$\begin{aligned}
Y &= \left(1 + \frac{r}{n}\right)^{n} - 1 \\
Y &= \left(1 + \frac{0.045}{12}\right)^{12} - 1 \\
Y &\approx 1.0459398 - 1 \\
Y &\approx 0.04594
\end{aligned}$$

The effective annual yield is approximately 4.59%.

6. Planning for Education: Present Value. Parents wish to have $\$120,000$ available for a child's education. If the child is now 1 year old, how much money must be set aside at 8% compounded monthly to meet their financial goal when the child is 18?

Solution. To find the required initial investment, we use the Present Value formula:

$$P = \frac{A}{(1 + \frac{r}{n})^{nt}}$$

where:

- $A = 120,000$ (The target goal)

- $r = 0.08$ (Annual interest rate)
- $n = 12$ (Monthly compounding)
- $t = 17$ (Years remaining until the child is 18: $18 - 1 = 17$)

Substituting these values into the formula:

$$P = \frac{120,000}{(1 + \frac{0.08}{12})^{12\times17}}$$
$$P = \frac{120,000}{(1.0066667)^{204}}$$
$$P \approx \frac{120,000}{3.876778}$$
$$P \approx 30,953.54$$

The parents must set aside \$30,953.54 today to meet the \$120,000 goal in 17 years.

7. **The Historical DeHaven Debt Calculation.** In 1777, Jacob DeHaven loaned George Washington's army \$450,000 in gold and supplies. Due to a disagreement over repayment, he was never repaid. In 1989, his descendants sued the U.S. government over the 212-year-old debt. If the DeHavens used an interest rate of 6% and daily compounding, how much money did the DeHaven family demand in their lawsuit?

Solution. To find the total value of the debt after 212 years, we use the compound interest formula:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

where:

- $P = 450,000$ (Initial loan)
- $r = 0.06$ (Annual interest rate)
- $n = 365$ (Daily compounding)
- $t = 212$ (Years from 1777 to 1989)

Substituting these values:

$$A = 450,000\left(1 + \frac{0.06}{365}\right)^{365\times212}$$
$$A = 450,000(1.00016438)^{77,380}$$
$$A \approx 450,000(336,543.15)$$
$$A \approx 151,444,417,500$$

The DeHaven family's demand in the lawsuit was approximately \$151.4 billion. (Depending on rounding, specific historical claims often cited amounts ranging from \$141 to \$151 billion).

Saving Money: Annuities

An **annuity** is a sequence of equal (regular) payments made into an account over time periods. An IRA is an example of an annuity. The value of an annuity is the sum of all deposits plus all interest paid.

Example 3.21 Annuity Basics. Suppose you deposit $1,000 into a savings plan at the end of each year for three years. The interest rate is 8% per year compounded annually.

1 Find the value of the annuity after three years.

2 Find the interest earned.

Solution. We calculate the balance step-by-step:

- Value at the end of year 1: $1,000
- Value at the end of year 2: $1,000(1+0.08)+1,000=\$2,080$
- Value at the end of year 3: $2,080(1+0.08)+1,000=\$3,246.40$

The value of the annuity at the end of three years is $3,246.40. Since you made three payments of $1,000, the total principal deposited is $3,000.

The interest is:

$$I = 3,246.40 - 3,000 = \$246.40$$

▲

Definition 3.22 Value of an Annuity: Interest Compounded n Times per Year. If P is the deposit made at the end of each compounding period for an annuity that pays an annual interest rate r compounded n times per year, the future value, FV or A, of the annuity after t years is:

$$FV = A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}{\left(\frac{r}{n}\right)} \tag{3.3}$$

◆

Example 3.23 Retirement Savings: IRA Annuity. To save for retirement, you decide to deposit $1,000 into an IRA at the end of each year for the next 30 years. If you can count on an interest rate of 10% per year compounded annually:

1 How much will you have from the IRA after 30 years?

2 Find the interest.

Solution. We use the future value formula for an annuity:

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}{\left(\frac{r}{n}\right)}$$

where $P = 1,000$, $r = 0.10$, $n = 1$, and $t = 30$.

1 Substituting the values into the formula:

$$A = \frac{1,000\left[(1+0.10)^{30}-1\right]}{0.10}$$

$$A = \frac{1,000\left[(1.1)^{30} - 1\right]}{0.10}$$
$$A \approx \frac{1,000(17.4494 - 1)}{0.10}$$
$$A \approx 164,494.02$$

After 30 years, the IRA will have $\$164,494.02$.

2 To find the interest, we subtract the total deposits from the future value. Total deposits $= 1,000 \times 30 = \$30,000$.

$$I = 164,494.02 - 30,000$$
$$I = 134,494.02$$

The total interest earned is $\$134,494.02$.

▲

Example 3.24 Retirement Savings with Monthly Compounding. At age 25, to save for retirement, you decide to deposit $200 at the end of each month into an IRA that pays 7.5% compounded monthly.

1 How much will you have from the IRA when you retire at age 65?

2 Find the interest.

Solution. We use the future value formula for an annuity:

$$A = \frac{P\left[\left(1 + \frac{r}{n}\right)^{nt} - 1\right]}{\left(\frac{r}{n}\right)}$$

where:

- $P = 200$ (Monthly deposit)
- $r = 0.075$ (Annual interest rate)
- $n = 12$ (Compounding periods per year)
- $t = 40$ (Years of investing: $65 - 25 = 40$)

1 Substituting the values into the formula:

$$A = \frac{200\left[\left(1 + \frac{0.075}{12}\right)^{12\times 40} - 1\right]}{\left(\frac{0.075}{12}\right)}$$
$$A = \frac{200\left[(1.00625)^{480} - 1\right]}{0.00625}$$
$$A \approx \frac{200(19.92053 - 1)}{0.00625}$$
$$A \approx 605,457.09$$

At age 65, the IRA will be worth $\$605,457.09$.

2 To find the interest, subtract the total deposits from the future value. Total deposits $= 200 \times 12 \times 40 = \$96,000$.

$$I = 605,457.09 - 96,000$$
$$I = 509,457.09$$

The total interest earned is $\$509,457.09$.

▲

Checkpoint 3.25 Calculating Future Value of an Annuity. To save for a down payment on a home, you deposit \$500 at the end of each quarter into an account that pays 4% compounded quarterly. How much will be in the account after 6 years?

Solution. We use the future value formula for an annuity:

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}{\left(\frac{r}{n}\right)}$$

where:

- $P = 500$ (Quarterly deposit)
- $r = 0.04$ (Annual interest rate)
- $n = 4$ (Compounded quarterly)
- $t = 6$ (Number of years)

Substituting these values into the formula:

$$A = \frac{500\left[\left(1+\frac{0.04}{4}\right)^{4\times 6}-1\right]}{\left(\frac{0.04}{4}\right)}$$
$$A = \frac{500\left[(1.01)^{24}-1\right]}{0.01}$$
$$A \approx \frac{500(1.26973-1)}{0.01}$$
$$A \approx 13,486.50$$

After 6 years, the account balance will be $\$13,486.50$.

Example 3.26 Logic Check: Annuities vs. Lump Sums. Determine whether each statement is True or False and explain your reasoning.

1 With the same interest rate, compounding period, and time period, an annuity will generate more interest than a lump sum deposit.

2 By putting \$20 at the end of each month into an IRA that pays 3.5% compounded monthly, I'll be able to retire comfortably in just 30 years.

Solution.

1 False. A lump sum deposit of the same total amount invested earns more interest because the entire amount is present at the beginning and earns interest for the full duration. In an annuity, the money is deposited gradually, so later payments have less time to grow.

2 False. We can calculate the total using the annuity formula where $P = 20$, $r = 0.035$, $n = 12$, and $t = 30$:

$$A = \frac{20\left[(1+\frac{0.035}{12})^{12\times 30}-1\right]}{\frac{0.035}{12}}$$
$$A \approx \$12,705.54$$

A total of $\$12,705.54$ is not nearly enough to sustain a comfortable retirement.

▲

Example 3.27 Sinking Fund: Monthly Savings for College. Parents of a baby girl are in a financial position to begin saving for her college education. They plan to have $\$100,000$ in a college fund in 18 years by making regular, end-of-month deposits in an annuity that pays 9% compounded monthly.

1 How much should they deposit each month?

2 How much of the goal comes from interest?

Solution. To find the required monthly payment (P), we use the sinking fund formula, which is the annuity formula solved for P:

$$P = \frac{A\left(\frac{r}{n}\right)}{\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}$$

where $A = 100,000$, $r = 0.09$, $n = 12$, and $t = 18$.

1 Substituting the values:

$$P = \frac{100,000\left(\frac{0.09}{12}\right)}{\left[\left(1+\frac{0.09}{12}\right)^{12\times 18}-1\right]}$$
$$P = \frac{750}{(1.0075)^{216}-1}$$
$$P \approx \frac{750}{5.022575-1}$$
$$P \approx 186.45$$

The parents should deposit $186.45 each month.

2 To find the interest, we first calculate the total principal deposited:

$$186.45 \times 12 \times 18 = \$40,273.20$$

Subtracting this from the future value:

$$I = 100,000 - 40,273.20 = \$59,726.80$$

The amount from interest is $\$59,726.80$.

▲

Checkpoint 3.28 Calculating the Value of an Annuity. You deposit $2,000 into a savings plan at the end of each year for three years. The interest rate is 10% per year compounded annually.

1 Find the value of the annuity after three years.

2 Find the interest earned.

Solution. We use the future value formula for an annuity:

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}{\left(\frac{r}{n}\right)}$$

where $P = 2000$, $r = 0.10$, $n = 1$, and $t = 3$.

1 Substituting the values into the formula:

$$\begin{aligned} A &= \frac{2000\left[(1+0.10)^3 - 1\right]}{0.10} \\ &= \frac{2000\left[(1.1)^3 - 1\right]}{0.10} \\ &= \frac{2000(1.331 - 1)}{0.10} \\ &= 6620 \end{aligned}$$

The value of the annuity after three years is \$6,620.

2 The total amount of deposits made is $3 \times 2000 = 6000$. The interest earned is the difference between the future value and the total deposits:

$$I = 6620 - 6000 = 620$$

The interest earned is \$620.

Checkpoint 3.29 Retirement Savings: Growth After Contributions Cease. To save for retirement, you decide to deposit \$3,000 into an IRA at the end of each year for the ***next 10*** years. If you can count on an interest rate of 9% per year compounded annually:

1 How much will you have from the IRA ***after 40*** years, assuming no additional deposits or withdrawals after the 10 years of contribution?

2 Find the interest earned.

Solution. This problem requires two steps. First, we find the value of the annuity after the 10-year contribution period. Second, we treat that amount as a lump sum that grows for the remaining 30 years.

Step 1: Future Value of the Annuity (Years 1-10)

$$\begin{aligned} A_{10} &= \frac{3000\left[(1+0.09)^{10} - 1\right]}{0.09} \\ &= \frac{3000(2.36736 - 1)}{0.09} \\ &\approx 45578.80 \end{aligned}$$

Step 2: Compound Interest on the Lump Sum (Years 11-40) The balance grows for $40 - 10 = 30$ more years using the formula $A = P(1+r)^t$:

$$\begin{aligned} A_{40} &= 45578.80(1+0.09)^{30} \\ &= 45578.80(13.26768) \\ &\approx 604725.10 \end{aligned}$$

1 After 40 years, the total value will be \$604,725.10.

2 The total deposits were $3000 \times 10 = 30000$.

$$I = 604725.10 - 30000 = 574725.10$$

The interest earned is \$574,725.10.

Exercises for Practice

1. Retirement Savings Analysis. At age 30 to save for retirement, you decide to deposit \$200 at the end of each month into an IRA that pays 13.8% compounded monthly.

1 How much will you have from the IRA when you retire at age 65?

2 Find the interest earned.

Solution. We use the future value formula for an ordinary annuity:

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}{\left(\frac{r}{n}\right)}$$

where:

- $P = 200$ (Monthly deposit)
- $r = 0.138$ (Annual interest rate)
- $n = 12$ (Compounding periods per year)
- $t = 35$ (Years of saving: $65 - 30 = 35$)

1 Substituting the values into the formula:

$$A = \frac{200\left[\left(1+\frac{0.138}{12}\right)^{12\times 35}-1\right]}{\left(\frac{0.138}{12}\right)}$$
$$A = \frac{200\left[(1.0115)^{420}-1\right]}{0.0115}$$
$$A \approx 2,100,992.95$$

You will have approximately $\$2,100,992.95$ in the IRA at age 65.

2 First, find the total amount of your deposits over 35 years:

$$200 \times 12 \times 35 = \$84,000$$

The interest earned is the future value minus these deposits:

$$2,100,992.95 - 84,000 = \$2,016,992.95$$

The total interest earned is $\$2,016,992.95$.

2. Calculating Monthly Deposits to Reach a Goal. How much should you deposit at the end of each month into an IRA that pays 6.5% compounded monthly to have \$2 million when you retire in 45 years? How much of the \$2 million comes from interest?

Solution. To find the required monthly payment (P), we use the sinking fund formula:

$$P = \frac{A\left(\frac{r}{n}\right)}{\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}$$

where:

- $A = 2,000,000$ (The target goal)

- $r = 0.065$ (Annual interest rate)
- $n = 12$ (Monthly compounding)
- $t = 45$ (Time in years)

Substituting these values into the formula:

$$\begin{aligned}
P &= \frac{2,000,000\left(\frac{0.065}{12}\right)}{\left[\left(1+\frac{0.065}{12}\right)^{12\times 45}-1\right]} \\
P &= \frac{10,833.333}{(1.0054167)^{540}-1} \\
P &\approx \frac{10,833.333}{21.21736-1} \\
P &\approx 510.59
\end{aligned}$$

The required monthly deposit is \$510.59.

To find the interest, we first calculate the total of all deposits:

$$510.59 \times 12 \times 45 = \$275,718.60$$

Subtracting this from the $\$2,000,000$ final value:

$$I = 2,000,000 - 275,718.60 = \$1,724,281.40$$

Over the 45 years, $\$1,724,281.40$ of the final balance comes from interest.

3. **Long-Term Retirement Savings Analysis.** At age 25, to save for retirement, you decide to deposit \$100 at the end of each month in an IRA that pays 6.25% compounded monthly. How much will you have from the IRA when you retire at age 70? What is the interest you earned?

Solution. We use the future value formula for an ordinary annuity:

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}{\left(\frac{r}{n}\right)}$$

where:

- $P = 100$ (Monthly deposit)
- $r = 0.0625$ (Annual interest rate)
- $n = 12$ (Compounding periods per year)
- $t = 45$ (Years of saving: $70 - 25 = 45$)

Substituting the values into the formula:

$$\begin{aligned}
A &= \frac{100\left[\left(1+\frac{0.0625}{12}\right)^{12\times 45}-1\right]}{\left(\frac{0.0625}{12}\right)} \\
A &= \frac{100\left[(1.0052083)^{540}-1\right]}{0.0052083} \\
A &\approx \frac{100(16.28236-1)}{0.0052083} \\
A &\approx 293,421.36
\end{aligned}$$

At age 70, you will have approximately $\$293,421.36$ in the IRA.

To find the interest earned, first calculate the total of your deposits:

$$100 \times 12 \times 45 = \$54,000$$

Subtracting the deposits from the future value:

$$I = 293,421.36 - 54,000 = \$239,421.36$$

The total interest earned is $\$239,421.36$.

4. **Quarterly Deposits to Reach a Retirement Goal.** How much should you deposit at the end of every 3 months into an IRA that pays 6% compounded quarterly to have \$1.5 million when you retire in 40 years? How much of the \$1.5 million comes from interest?

Solution. To find the required quarterly payment (P), we use the sinking fund formula:

$$P = \frac{A\left(\frac{r}{n}\right)}{\left[\left(1+\frac{r}{n}\right)^{nt} - 1\right]}$$

where:

- $A = 1,500,000$ (The target goal)
- $r = 0.06$ (Annual interest rate)
- $n = 4$ (Quarterly compounding)
- $t = 40$ (Time in years)

Substituting these values into the formula:

$$\begin{aligned}
P &= \frac{1,500,000\left(\frac{0.06}{4}\right)}{\left[\left(1+\frac{0.06}{4}\right)^{4\times 40} - 1\right]} \\
P &= \frac{22,500}{(1.015)^{160} - 1} \\
P &\approx \frac{22,500}{10.85055 - 1.85055} \\
P &\approx 2,284.14
\end{aligned}$$

The required quarterly deposit is $\$2,284.14$.

To find the interest, we first calculate the total of all deposits over 40 years:

$$2,284.14 \times 4 \times 40 = \$365,462.40$$

Subtracting this from the $\$1,500,000$ final value:

$$I = 1,500,000 - 365,462.40 = \$1,134,537.60$$

Over the 40 years, $\$1,134,537.60$ of the final balance comes from interest.

Borrowing Money: Mortgage

A ***mortgage*** is a long-term loan (perhaps up to 15 or 30 years) for the purpose of buying a home, and for which the property is pledged as security for payment.

The ***down payment*** is the portion of the sale price of the home that the buyer initially pays to the seller.

The minimum required down payment is computes as a percentage of the sale price.

The ***amount of the mortgage*** is the difference between the sale price and the down payment.

The ***points*** are charged for your mortgage. One point equals one percent of the loan amount.

For example, you make a 10% down payment of \$120,000 home, the amount of the mortgage is $\$120,000 - \$12,000 = \$108,000$. So, if your mortgage is \$108,000, one point is \$1,080. You pay these points up front when you begin the mortgage.

Many financial institutions or mortgage companies provide mortgage loans. Some companies, called ***mortgage broker*** offer to find you a mortgage lender willing to make you a loan. Mortgages can have a fixed interest rate or a variable interest rate. ***Fixed-rate mortgages*** have same monthly payment during the entire time of the loan.

Definition 3.30 Loan Payment Formula for Fixed Installment Loans. The regular payment amount, PMT, required to repay a loan of P dollars paid n times per year over t years at an annual rate r is given by:

$$PMT = \frac{P\left(\frac{r}{n}\right)}{\left[1-\left(1+\frac{r}{n}\right)^{-nt}\right]} \tag{3.4}$$

where:

P	is the principal amount (loan balance)
r	is the annual interest rate (as a decimal)
n	is the number of payments per year
t	is the total number of years

◆

Example 3.31 . The price of a home is \$195,000. The bank requires a 10% down payment and two points at the time of closing. The cost of the home is financed with 30-year fixed rate mortgage at 7.5%.

1. Find the required down payment.
2. Find the amount of the mortgage.
3. How much must be paid for the two points at closing?
4. Find the monthly payment
5. Find the total interest paid over 30 years.

Solution.

1. The required down payment is 10% of $195,000 or

$$0.10 \times \$195,000 = \$19,500.$$

2. The amount of the mortgage is the difference between the price of the home and the down payment:

$$\text{Amount of mortgage} = \$195,000 - \$19,500 = \$175,500.$$

3. To find the cost of two points on a mortgage of $175,500:

$$2 \text{ points} = 2\% \text{ of mortgage} = 0.02 \times \$175,500 = \$3,510.$$

The down payment ($19,500) is paid to the seller and the cost of two points ($3,510) is paid to the lending institution.

Computing Monthly Mortgage Payments. The monthly mortgage payment, PMT, required to repay a loan of P dollars paid n times per year over t years at an annual rate r is:

$$PMT = \frac{P\left(\frac{r}{n}\right)}{\left[1 - \left(1 + \frac{r}{n}\right)^{-nt}\right]}$$

4. The monthly payments are:

$$PMT = \frac{175500\left(\frac{0.075}{12}\right)}{\left[1 - \left(1 + \frac{0.075}{12}\right)^{-12\cdot 30}\right]} \approx \$1,227.12$$

5. The total payments are calculated by multiplying the monthly payment by the total number of periods (360 months):

$$\begin{aligned}\text{Total Payments} &= \text{monthly payments} \times \text{month} \times \text{years}\\ &= \$1,227.12 \times 360 = \$441,763.20.\end{aligned}$$

The interest paid over 30 years is the total payments minus the original principal:

$$\begin{aligned}\text{Interest} &= \text{total payments} - \text{amount of mortgage}\\ &= \$441,763.20 - \$175,500 = \$266,263.20.\end{aligned}$$

▲

Example 3.32 Comparing 30-Year and 15-Year Mortgages. In Example 3.31,a mortgage of $175,500 was financed with a 30-year fixed rate at 7.5%.

1. Find the monthly payment for the 30-year mortgage.
2. Find the total interest paid over 30 years.
3. Find the total interest paid if the loan was instead for 15 years at the same rate.
4. How much interest is saved by reducing the mortgage from 30 to 15 years?

Solution.

1. We use the formula

$$PMT = \frac{P(r/n)}{1-(1+r/n)^{-nt}}$$

with $P = 175500$, $r = 0.075$, $n = 12$, and $t = 30$:

$$PMT = \frac{175500(0.075/12)}{1-(1+0.075/12)^{-12\cdot 30}} \approx \$1,227.12$$

2. The total interest for 30 years is the total payments minus the principal:

$$\text{Total Payments} = \$1,227.12 \times 360 = \$441,763.20$$

$$\text{Interest}_{30} = \$441,763.20 - \$175,500 = \$266,263.20$$

3. First, find the 15-year monthly payment ($t = 15$):

$$PMT = \frac{175500(0.075/12)}{1-(1+0.075/12)^{-12\cdot 15}} \approx \$1,626.91$$

Now calculate the total 15-year interest:

$$\text{Total Payments} = \$1,626.91 \times 180 = \$292,843.80$$

$$\text{Interest}_{15} = \$292,843.80 - \$175,500 = \$117,343.80$$

4. The interest saved is the difference between the two interest totals:

$$\text{Savings} = \$266,263.20 - \$117,343.80 = \$148,919.40$$

▲

Example 3.33 Condominium Mortgage Analysis. The price of a condominium is $280,000. The bank requires a 5% down payment and one point at the time of closing. The cost of the condominium is financed with a 30-year fixed-rate mortgage at 8%.

1. Find the required down payment.
2. Find the amount of the mortgage.
3. How much must be paid for the one point at closing?
4. Find the monthly payment by using the monthly payment formula.
5. Find the total cost of interest over 30 years.

Solution.

1. The down payment is 5% of the purchase price:

$$0.05 \times \$280,000 = \$14,000$$

2. The mortgage amount is the price minus the down payment:

$$\$280,000 - \$14,000 = \$266,000$$

3. One point is 1% of the mortgage amount:

$$0.01 \times \$266,000 = \$2,660$$

4. Using the formula
$$PMT = \frac{P(r/n)}{1-(1+r/n)^{-nt}}$$
with $P = 266,000$, $r = 0.08$, $n = 12$, and $t = 30$:
$$PMT = \frac{266000(0.08/12)}{1-(1+0.08/12)^{-360}} \approx \$1,951.81$$

5. Total payments over 30 years are
$$\$1,951.81 \times 360 = \$702,651.60.$$
The total interest is:
$$\text{Total Interest} = \$702,651.60 - \$266,000 = \$436,651.60$$

▲

Checkpoint 3.34 Student Loan Term Comparison. A student graduates from college with a loan of \$36,000 at an 8% annual interest rate.

1. Find the monthly payments and the total interest for a 10-year loan term.
2. If the interest rate remains at 8% and the loan term is reduced to 5 years, how much more must the student pay each month and how much less will be paid in total interest?

Answer.

1. Monthly payment: \$436.78; Total interest: \$16,413.60
2. Monthly increase: \$293.07; Interest saved: \$8,414.40

Solution.

1. For $t = 10$:
$$PMT = \frac{36000(0.08/12)}{1-(1+0.08/12)^{-120}} \approx \$436.78$$
$$\text{Total Interest} = (\$436.78 \times 120) - \$36,000 = \$16,413.60$$

2. For $t = 5$:
$$PMT = \frac{36000(0.08/12)}{1-(1+0.08/12)^{-60}} \approx \$729.85$$
$$\text{Total Interest} = (\$729.85 \times 60) - \$36,000 = \$7,791.00$$
The monthly increase is $\$729.85 - \$436.78 = \$293.07$. The interest saved is $\$16,413.60 - \$7,791.00 = \$8,422.60$. (Note: Small differences may occur due to rounding).

Checkpoint 3.35 Mortgage Comparison with Points and Closing Costs. Which mortgage loan has the greater total cost and by how much? Consider a \$210,000 mortgage with these two options:

- **Mortgage A**: 20-year fixed at 7% with closing costs of \$3,000 and 2 points.
- **Mortgage B**: 20-year fixed at 6.75% with closing costs of \$1,750 and 4

points.

Answer. Mortgage A has the greater total cost by $4,577.20.

Solution. The total cost for each mortgage includes the sum of all monthly payments, the cost of points, and the fixed closing costs.

1. *Mortgage A Calculations.*

   First, calculate the monthly payment (PMT):

   $$PMT = \frac{210000(0.07/12)}{1-(1+0.07/12)^{-240}} \approx \$1,628.13$$

   Next, find the total cost of payments, points, and closing:

   $$\begin{aligned} \text{Total Payments} &= \$1,628.13 \times 240 = \$390,751.20 \\ \text{Points Cost} &= 0.02 \times \$210,000 = \$4,200.00 \\ \text{Total Cost}_A &= \$390,751.20 + \$4,200.00 + \$3,000.00 = \$397,951.20 \end{aligned}$$

2. *Mortgage B Calculations.*

   First, calculate the monthly payment (PMT):

   $$PMT = \frac{210000(0.0675/12)}{1-(1+0.0675/12)^{-240}} \approx \$1,596.76$$

   Next, find the total cost:

   $$\begin{aligned} \text{Total Payments} &= \$1,596.76 \times 240 = \$383,222.40 \\ \text{Points Cost} &= 0.04 \times \$210,000 = \$8,400.00 \\ \text{Total Cost}_B &= \$383,222.40 + \$8,400.00 + \$1,750.00 = \$393,372.40 \end{aligned}$$

3. *Comparison.*

   Mortgage A is more expensive. The difference is:

   $$\$397,951.20 - \$393,372.40 = \$4,578.80$$

   (Note: Differences in final cents may vary based on rounding the monthly payment).

Definition 3.36 Maximum Purchase Price Formula. The ***maximum purchase price*** for a house is given by

$$\text{maximum purchase price} = \frac{PMT \cdot \left[1-\left(1+\frac{r}{n}\right)^{-nt}\right]}{\left(\frac{r}{n}\right)}$$

where:

$$\begin{aligned} PMT &\quad \text{is the monthly mortgage payment} \\ n &\quad \text{is the number of payments per year} \\ t &\quad \text{is the total number of years (loan term)} \\ r &\quad \text{is the annual interest rate (as a decimal)} \end{aligned}$$

◆

Example 3.37 Calculating Maximum Mortgage Amount. Amanda and Fred are buying a house on a 30-year mortgage. They can afford a monthly payment of $800. If the annual interest rate (APR) is 3.75%, what is the maximum mortgage amount they can take out?

Solution. We use the maximum purchase price formula with $PMT = 800$, $r = 0.0375$, $n = 12$, and $t = 30$:

$$\text{Max Price} = \frac{800 \cdot \left[1 - \left(1 + \frac{0.0375}{12}\right)^{-12 \cdot 30}\right]}{\left(\frac{0.0375}{12}\right)}$$

First, calculate the periodic interest rate: $\frac{0.0375}{12} = 0.003125$.
Now, substitute into the formula:

$$\begin{aligned}
\text{Max Price} &= \frac{800 \cdot \left[1 - (1.003125)^{-360}\right]}{0.003125} \\
\text{Max Price} &= \frac{800 \cdot [1 - 0.325016]}{0.003125} \\
\text{Max Price} &= \frac{800 \cdot 0.674984}{0.003125} \\
\text{Max Price} &\approx \$172,795.90
\end{aligned}$$

Amanda and Fred can take out a mortgage of approximately ***$172,795.90***.

▲

Checkpoint 3.38 Home Affordability and Loan Terms. Suppose you want to purchase a house and your take-home pay is $3,220 per month. You wish to stay within recommended guidelines by spending 1/4 of your take-home pay on a house payment. You have $15,300 saved for a down payment. With your excellent credit, you can get an APR of 3.28% compounded monthly.

1. What is the total cost of a house you could afford with a 15-year mortgage?
2. What is the most that you could afford with a traditional 30-year mortgage?

Answer.

1. 15-year mortgage: $130,175.76
2. 30-year mortgage: $199,444.60

Solution. First, calculate the allowed monthly payment (PMT):

$$PMT = \frac{1}{4} \times \$3,220 = \$805$$

The monthly interest rate is $\frac{r}{n} = \frac{0.0328}{12} \approx 0.0027333$.

1. *15-Year Mortgage Affordability.*

   Using the Maximum Purchase Price formula with $n = 12$ and $t = 15$ ($nt = 180$):

   $$P = \frac{805 \cdot [1 - (1 + 0.0027333)^{-180}]}{0.0027333} \approx \$114,875.76$$

   The total house cost includes the down payment:

   $$\text{Total Cost} = \$114,875.76 + \$15,300 = \$130,175.76$$

2. *30-Year Mortgage Affordability.*

   Using the formula with $t = 30$ ($nt = 360$):

$$P = \frac{805 \cdot [1 - (1 + 0.0027333)^{-360}]}{0.0027333} \approx \$184,144.60$$

   The total house cost includes the down payment:

$$\text{Total Cost} = \$184,144.60 + \$15,300 = \$199,444.60$$

Identify Formulas to use: Simple, Compound, Annuity, Loan

Identify the Formula to Use

1. Lonnie needs extra money to buy a truck to start up a delivery service. He takes out a simple interest loan for \$7,000 for 6 months at a rate of 7.25%. How much interest must he pay, and what is the future value of the loan?

Answer. $I = Prt$; Interest: \$253.75; Future Value: \$7,253.75

2. If you placed \$1 into an account that paid interest at a rate of 5% and compounded the interest daily, how much would that account is worth in 300 years?

Answer. $A = P(1 + r/n)^{nt}$; \$3,268,864.51

3. You are purchasing a house for \$155,000. The bank requires a down payment of 15% and 2.25 points at closing. The bank will give you a 20 year mortgage at 3.8%. What is the monthly payment you will make? What is the total interest paid?

Answer. $PMT = \frac{P(r/n)}{1-(1+r/n)^{-nt}}$; Monthly Payment: \$783.50; Total Interest: \$56,289.44

4. You deposit \$8,500 for 9 years in an account that earns 2.5% interest and compounds weekly. How much interest will you earn?

Answer. $A = P(1 + r/n)^{nt}$; \$2,143.91

5. You lend 650 dollars to your younger brother for tuition. He agrees to pay you back \$720 in 4 months. What is the interest rate he paid you?

Answer. $r = \frac{I}{Pt}$; 32.31%

6. A mother invests \$ 2000 in a bank account at the time of her daughter's birth. The interest is compounded quarterly at a rate of 8%. What will be the value of the daughter's account on her twentieth birthday?

Answer. $A = P(1 + r/n)^{nt}$; \$9,750.88

7. Your grandmother lent you \$2000 for 4 years to buy a car. You agreed to pay her 5% interest. How much do you have to pay her in total at the end of 4 years?

Answer. $A = P(1 + rt)$; \$2,400.00

8. Which is the better choice: \$1000 deposited for a year at a rate of 5.5% compounded semiannually or at a rate of 5.4% compounded quarterly?

Answer. $APY = (1 + r/n)^n - 1$; 5.5% compounded semiannually (\$1,055.76 vs \$1,055.10)

9. Ellen puts $100 each month into an IRA that pays 7.4% annually. How much money will she have in 30 years for her retirement? How much will come from interest?

Answer. $A = \frac{PMT[(1+r/n)^{nt}-1]}{r/n}$; Total Value: $132,410.87; Interest: $96,410.87

10. You want to save for your three year olds college education. You decide that you need $200,000 in 12 years. How much do you need to deposit monthly into an account that pays 8.7% interest and compounds monthly.

Answer. $PMT = \frac{A(r/n)}{(1+r/n)^{nt}-1}$; $796.86

11. Joe and Susan are purchasing a condo for $350,000. The bank requires a down payment of 30% and 4.75 points at closing. You are considering a 15 year mortgage at 4.75% and a 30 year mortgage at 3.2%. Which mortgage will result in a lower monthly payment?

Answer. $PMT = \frac{P(r/n)}{1-(1+r/n)^{-nt}}$; Option B: 30-year at 3.2% ($1,061.27 vs Option A: $1,906.18)

12. Pete wants to buy a boat in 4 years. The boat costs $100,000. How much money does he need to deposit weekly into an account that pays 3.5% interest compounded weekly?

Answer. $PMT = \frac{A(r/n)}{(1+r/n)^{nt}-1}$; $447.93

13. How much money should be deposited today in an account that earns 10% compounded semiannually so that it will accumulate to $ 1000 in 12 years?

Answer. $P = \frac{A}{(1+r/n)^{nt}}$; $310.07

14. Juan can save $50 every week for his honeymoon. If Juan is getting married in two years how much money will he have spent on the trip if he puts it into an annuity that pays 6%?

Answer. $A = \frac{PMT[(1+r/n)^{nt}-1]}{r/n}$; $5,516.48

15. How much money should be deposited today in an account that earns 8% compounded quarterly so that it will accumulate to $ 3800 in 12 years?

Answer. $P = \frac{A}{(1+r/n)^{nt}}$; $1,468.32

Chapter Exercise

Exercises

1. Suppose a local sales tax rate is 4.5% and you purchase a car for $\$28,500$.

a How much sales tax is paid?

b How much is the total cost of the car?

Answer.

a $\$1,282.50$

b $\$29,782.50$

Solution.

a Tax $= 28500 \times 0.045 = 1282.50$.

b Total $= 28500 + 1282.50 = 29782.50$.

2. A bank offers a CD that pays a simple interest rate of 5.25%. How much must you put in this CD now in order to have $\$5,000$ for college tuition in five years?

Answer. $\$3,960.40$

Solution. Use $A = P(1+rt)$. $5000 = P(1+0.0525\times 5)$ $5000 = P(1.2625)$ $P = 5000/1.2625 \approx 3960.40$.

3. How much money should be deposited today in an account that earns 5% compounded quarterly so that it will accumulate to $\$15,000$ in ten years?

Answer. $\$9,126.10$

Solution. Use $P = A(1 + r/n)^{-nt}$. $P = 15000(1 + 0.05/4)^{-40}$ $P = 15000(1.0125)^{-40} \approx 9126.10$.

4. The cost of washer-dryer is $\$1,100$. We can finance this by paying $100 down and $110 per month for 12 months. Determine the finance charge.

Answer. $320

Solution. Total paid = Down payment + (Monthly payment $\times$ months) Total $= 100 + (110 \times 12) = 100 + 1320 = 1420$. Finance Charge $= 1420 - 1100 = 320$.

5. At the time of her grandson's birth, a grandmother deposits $\$5,000$ in an account that pays 6% compounded monthly. What will be the value of the account at the child's twenty-first birthday?

Answer. $\$17,560.10$

Solution. $A = 5000(1 + 0.06/12)^{(12\times 21)}$ $A = 5000(1.005)^{252} \approx 17560.10$.

6. You deposit $\$7,500$ in an account that pays 4.75% interest compounded monthly. Find the future value after one year and the effective annual yield.

Answer. FV: $\$7,864.08$; Yield: 4.85%

Solution. FV $= 7500(1 + 0.0475/12)^{12} \approx 7864.08$. Yield $= (1 + 0.0475/12)^{12} - 1 \approx 0.0485$ or 4.85%.

7. Find the value of an annuity of $\$1,000$ at the end of every 6 months at 4.5% compounded semiannually for 10 years.

Answer. Value: $\$24,911.52$; Interest: $\$4,911.52$

Solution. Use $A = P\frac{(1+r/n)^{nt}-1}{r/n}$. $A = 1000\frac{(1+0.045/2)^{20}-1}{0.045/2} \approx 24911.52$. Total deposits $= 1000\times 20 = 20000$. Interest $= 24911.52 - 20000 = 4911.52$.

8. Financial goal of $\$150,000$ at 4.75% compounded quarterly for 35 years. Find the periodic deposit.

Answer. Deposit: \$168.32

Solution. $P = \frac{A(r/n)}{(1+r/n)^{nt}-1}$ $P = \frac{150000(0.0475/4)}{(1+0.0475/4)^{140}-1} \approx 168.32$.

9. Loan of $\$36,000$ at 8% for 10 years.

a Find monthly payments and total interest.

b Compare to a 5-year term.

Answer.

a Payment: \$436.78; Interest: $\$16,413.60$

b Pay \$293.10 more per month; Save $\$8,620.80$ interest.

Solution. Use $PMT = \frac{Pr/n}{1-(1+r/n)^{-nt}}$.
10-yr: $PMT \approx 436.78$. Total paid $= 52413.60$. Interest $= 16413.60$.
5-yr: $PMT \approx 729.88$. Total paid $= 43792.80$. Interest $= 7792.80$.

Chapter 4

Hands-On Sessions

The following notes are intended to remind you what we covered each week. They are not a substitute for attending class and not a substitute for reading the textbook.

Chapter 5

In-Class Activities

These are copies of the in-class activities distributed during the semester.

Introduction Activity

This is the introduction to the activity.

1. This is the first exercise.

Chapter 6

Handouts

Chapter 7

Homework